Agentic Data Analysis Assistant

Turning Data into Actionable Insights

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Problem Statement

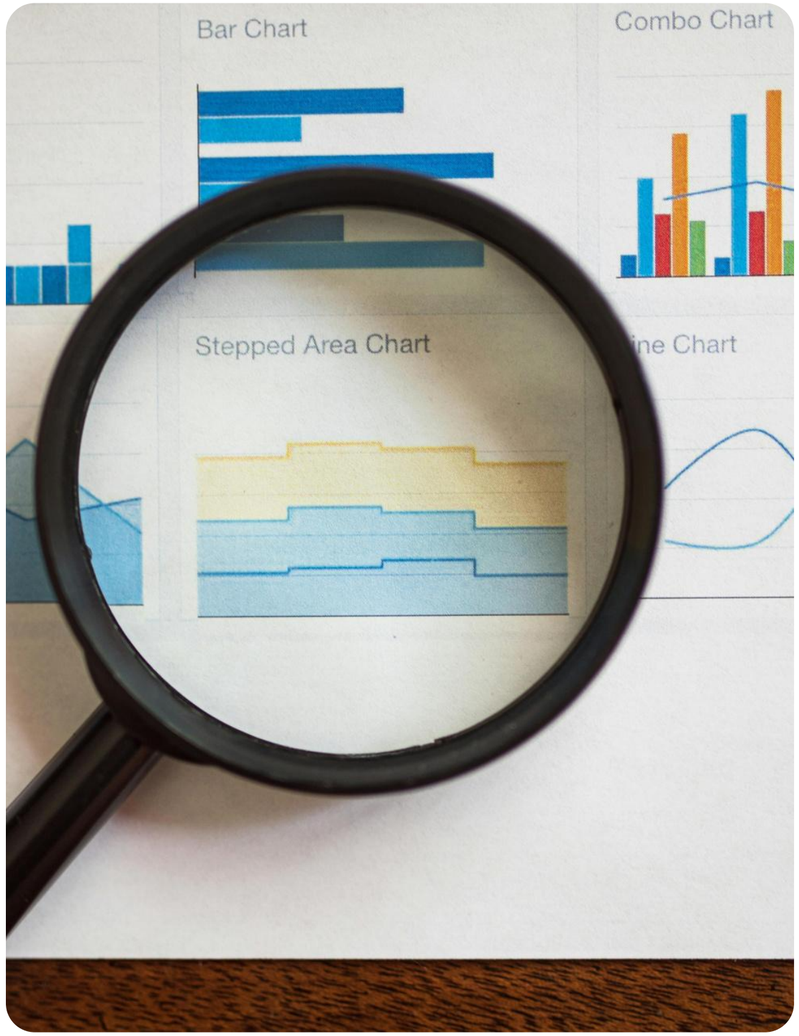
- Analysing CSV datasets manually requires programming knowledge and significant time.
- Users often struggle to identify appropriate statistical methods and visualisations.
- Traditional data-analysis workflows require users to manually perform multiple operations.
- Unrestricted AI-generated code execution can introduce security and reliability risks.

Our Proposed Solution

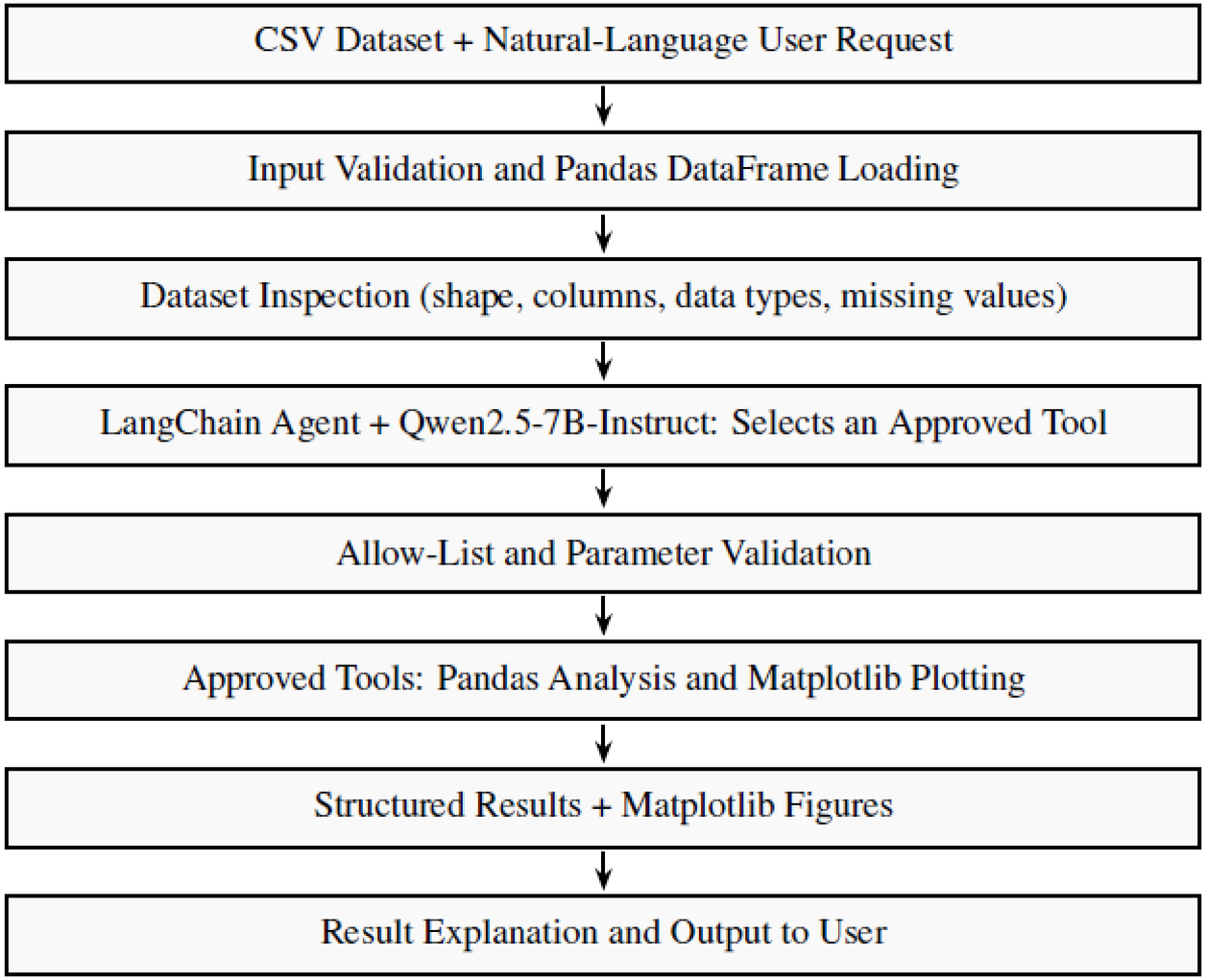
Develop an Agentic Data Analysis Assistant that uses an LLM to understand user requests, select appropriate analysis tools, calculate statistics using Pandas, and generate visualisations using Matplotlib.

Project Objectives

- Automated Dataset Inspection
- Intelligent Tool Selection
- Statistical Analysis
- Data Visualisation
- Safe and Controlled Execution



Proposed processing pipeline



Models and Technologies Used

Large Language Model (LLM)

- Qwen2.5-7B-Instruct
- Understands natural-language user requests.
- Helps select appropriate analysis tools.

Agent Framework

- LangChain
- Connects the language model with predefined tools.
- Manages tool selection and execution flow.

Data Analysis and Visualisation

- Python – Core programming language.
- Pandas – Dataset loading, inspection and statistical analysis.
- NumPy – Numerical operations.
- Matplotlib – Graph and chart generation.

Development Environment and Integration

- Google Colab – Development and testing environment.
- Hugging Face Inference Providers – Access to the language model.
- Google Colab Secrets – Secure storage of the Hugging Face API token.

Implementation Methodology

1. Dataset Loading

Accept CSV files from the user • Validate the uploaded file
Load data into a Pandas DataFrame

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2. Dataset Inspection

Identify rows and columns • Check data types and missing values
Understand the dataset structure

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3. Agent and Tool Integration

Connect Qwen2.5-7B-Instruct with LangChain • Interpret natural-language queries
Select the appropriate predefined tool

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4. Controlled Tool Execution

Validate tool names and input parameters
Allow execution only through approved operations • Prevent arbitrary code execution

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5. Analysis and Output

Perform statistical analysis using Pandas and NumPy • Generate charts using Matplotlib
Return results with natural-language explanations

Live Demonstration of the Agentic Data Analysis Assistant

1. **Upload Dataset**
2. **Dataset Inspection**
3. **Enter Natural-Language Query**
4. **Agent Processing**
5. **Display Results**



Results

1. Automated CSV dataset inspection
2. Natural-language-based analysis requests.
3. Descriptive statistical calculations using Pandas.
4. Data visualisations using Matplotlib.
5. Easy-to-understand explanations of analytical results

Usability and Limitations

Usability

- Reduces manual effort in data analysis.
- Makes data analysis easier for non-programmers.
- Allows users to request analysis using simple language.
- Provides statistics and visualisations in one workflow.

Limitations

- Requires a working connection to the Hugging Face inference service.
- Performance depends on the language model's ability to interpret queries correctly.
- Currently supports CSV datasets and predefined analysis tools.
- Large datasets may require additional processing time.
- Incorrect or unsupported user requests may need clarification.

Group Contributions

Group Members		
Anjali Kumari	2304008	Frontend development using Streamlit, dashboard design, dataset visualization, data exploration and frontend-backend integration.
Ayush Raj	2304018	Backend development, LLM integration, LangGraph agent workflow and analytical tool implementation.
Ankit Dash	2304009	Project research, testing, documentation, result analysis and report preparation.

Conclusion

The Agentic Data Analysis Assistant aims to simplify the data analysis process by allowing users to interact with CSV datasets through natural-language queries instead of writing code manually. By combining Qwen2.5-7B-Instruct and LangChain with Python libraries such as Pandas, NumPy and Matplotlib, the system is designed to automate dataset inspection, statistical analysis and data visualisation. It uses predefined tools and input validation to ensure controlled and safer execution. Overall, this project demonstrates how AI agents can make data analysis faster, easier and more accessible to users, including those without advanced programming knowledge.



Thank You